测不准关系

（一）测不准关系

（1）引

由上节讨论表明，两力学量算符对易则同时有确定值；若不对易，一般来说，不存在共同本征函数，不同时具有确定值。

问题：

两个不对易算符所对应的力学量在某一状态中究竟不确定到什么程度？即不确定度是多少？

测不准关系

（一）测不准的物理图像

（二）测不准关系的严格推导

（三）坐标和动量的测不准关系

（四）角动量的测不准关系

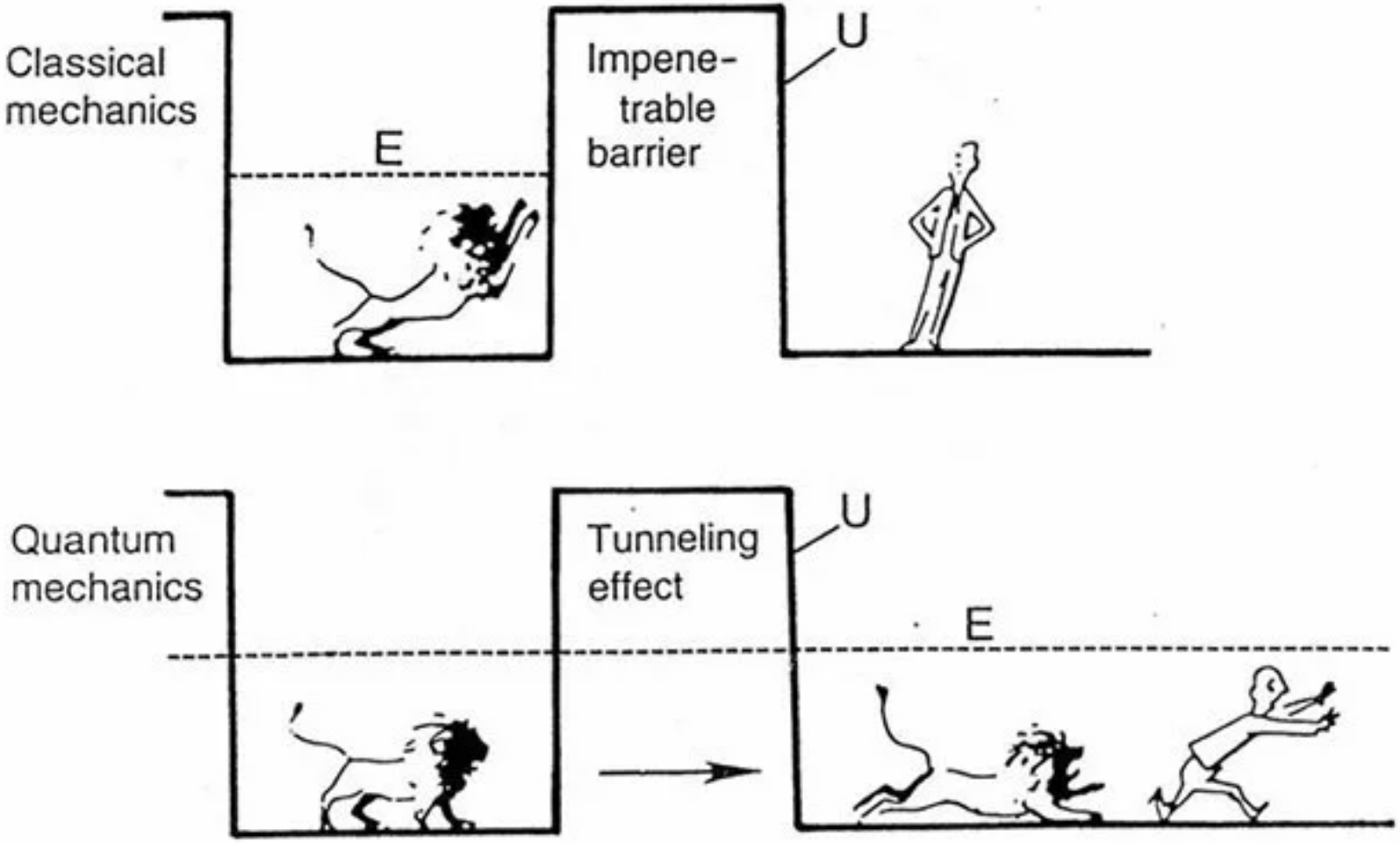
测不准关系

$$x \quad \Longleftrightarrow \quad p_x$$



测不准的物理图像是什么

测不准关系


Classical
mechanics
E
Impene-
trable
barrier
U
Quantum
mechanics
Tunneling
effect
U
E

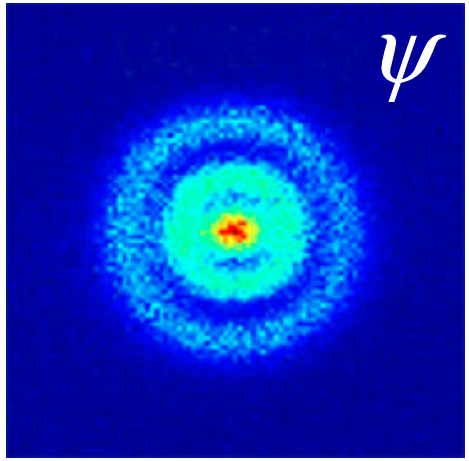
测不准关系

$$[\hat{L}_x, \hat{L}_y] = i\hbar \hat{L}_z$$

测不准关系

单个力学量的不确定度

an excited hydrogen atom



测量氢原子中电子的位置

	1	2	3	4	5	6	7	8	9	…
r(埃)	1	3	2	100	50	2	3	11	90	…

$< r >$=10埃

每次测量的偏差

$r - < r >$

测不准关系

不确定度：　测量值 $\mathbf{F_n}$ 与期望值 $<\mathbf{F}>$ 的偏差的大小。

定义算符

$$\Delta\hat{F} = \hat{F} - \overline{F}$$

若$\hat{F}$为厄密算符，则偏差$\Delta\hat{F} = \hat{F} - \overline{F}$仍为厄密算符。

证：

$$(\Delta\hat{F})^+ = (\hat{F} - \overline{F})^+ = \hat{F}^+ - \overline{F}^* = \hat{F} - \overline{F} = \Delta\hat{F}$$

测不准关系

II 测不准关系的严格推导

设二厄密算符对易关系为：$\hat{F}\hat{G}-\hat{G}\hat{F}=i\hat{k}$

是算符或普通数

为求二量不确定度 $\Delta\hat{F}$、$\Delta\hat{G}$,引入实参量 ξ 的辅助积分：

$$I(\xi)=\int |\xi\Delta\hat{F}\psi-i\Delta\hat{G}\psi|^2\,d\tau \geq 0$$

$$=\int (\xi\Delta\hat{F}\psi-i\Delta\hat{G}\psi)^*(\xi\Delta\hat{F}\psi-i\Delta\hat{G}\psi)d\tau$$

$$=\int [\xi(\Delta\hat{F}\psi)^*+i(\Delta\hat{G}\psi)^*](\xi\Delta\hat{F}\psi-i\Delta\hat{G}\psi)d\tau$$

$$=\xi^2\int (\Delta\hat{F}\psi)^*(\Delta\hat{F}\psi)d\tau-i\xi\int (\Delta\hat{F}\psi)^*(\Delta\hat{G}\psi)d\tau$$

$$+i\xi\int (\Delta\hat{G}\psi)^*(\Delta\hat{F}\psi)d\tau+\int (\Delta\hat{G}\psi)^*(\Delta\hat{G}\psi)d\tau$$

利用厄密算符的定义

$$=\xi^2\int \psi^*\Delta\hat{F}(\Delta\hat{F}\psi)d\tau-i\xi\int \psi^*\Delta\hat{F}(\Delta\hat{G}\psi)d\tau$$

$$+i\xi\int \psi^*\Delta\hat{G}(\Delta\hat{F}\psi)d\tau+\int \psi^*\Delta\hat{G}(\Delta\hat{G}\psi)d\tau$$

$$=\xi^2\int \psi^*(\Delta\hat{F})^2\psi d\tau-i\xi\int \psi^*(\Delta\hat{F}\Delta\hat{G}-\Delta\hat{G}\Delta\hat{F})\psi d\tau+\int \psi^*(\Delta\hat{G})^2\psi d\tau$$

测不准关系

$$\Delta\hat{F}\Delta\hat{G}-\Delta\hat{G}\Delta\hat{F}\equiv[\Delta\hat{F},\Delta\hat{G}]$$

$$=[\hat{F}-\bar{F},\ \hat{G}-\bar{G}]$$

常数

$$=[\hat{F}-\bar{F},\hat{G}]-[\hat{F}-\bar{F},\bar{G}]$$

常数

$$=[\hat{F},\hat{G}]-[\bar{F},\hat{G}]$$

$$=[\hat{F},\hat{G}]=i\hat{k}$$

$$0\le I(\xi)=\xi^2\int\psi^*(\Delta\hat{F})^2\psi d\tau-i\xi\int\psi^*(\Delta\hat{F}\Delta\hat{G}-\Delta\hat{G}\Delta\hat{F})\psi d\tau+\int\psi^*(\Delta\hat{G})^2\psi d\tau$$

$$=\xi^2\int\psi^*(\Delta\hat{F})^2\psi d\tau-i\xi\int\psi^*(i\hat{k})\psi d\tau+\int\psi^*(\Delta\hat{G})^2\psi d\tau$$

整理

$$=\xi^2\int\psi^*(\Delta\hat{F})^2\psi d\tau+\xi\int\psi^*\hat{k}\psi d\tau+\int\psi^*(\Delta\hat{G})^2\psi d\tau$$

$$=\xi^2\overline{(\Delta\hat{F})^2}+\xi\bar{k}+\overline{(\Delta\hat{G})^2}$$

测不准关系

$$I(\xi)=\xi^2\overline{(\Delta\hat{F})^2}+\xi\bar{k}+\overline{(\Delta\hat{G})^2}\geq 0$$

对任意实数ξ均成立

由代数二次式理论可知，该不等式成立的条件是系数必须满足下列关系：

$$\overline{(\Delta\hat{F})^2}\cdot\overline{(\Delta\hat{G})^2}\geq\frac{(\bar{k})^2}{4}$$

其中：$\bar{k}=\int\psi^*\hat{k}\psi d\tau$

均方偏差

$$\overline{(\Delta\hat{F})^2}=\overline{(\hat{F}-\bar{F})^2}=\overline{\hat{F}^2-2\hat{F}\bar{F}+\bar{F}^2}$$

$$=\overline{F^2}-\overline{2\hat{F}\bar{F}}+\overline{\bar{F}^2}=\overline{F^2}-2\bar{F}\bar{F}+\bar{F}^2$$

$$=\overline{F^2}-\bar{F}^2$$

两个不对易算符均方偏差关系式

测不准关系

测不准关系

（二）坐标和动量的测不准关系

（1）测不准关系

$$[\hat{F},\hat{G}]=i\hat{k} \qquad \overline{(\Delta\hat{F})^2}\cdot\overline{(\Delta\hat{G})^2}\geq\frac{(\bar{k})^2}{4}$$

$$\because[x,\hat{p}_x]=i\hbar \qquad \therefore\overline{(\Delta x)^2}\cdot\overline{(\Delta p_x)^2}\geq\frac{\hbar^2}{4}$$

表明：坐标与动量的均方偏差不能同时为零，其一越小，另一就越大。

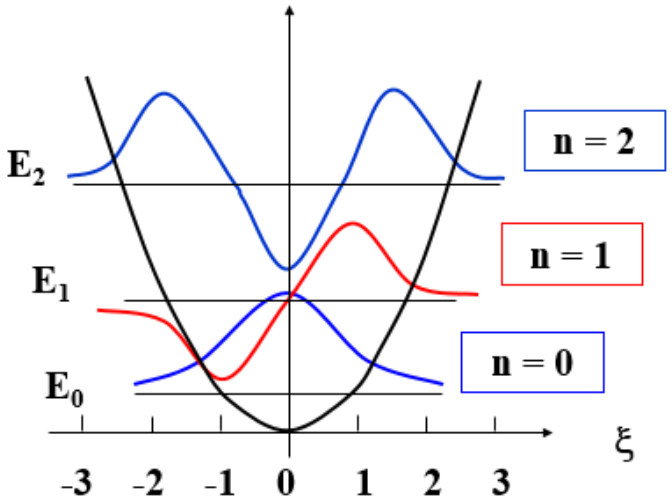
测不准关系

（2）线性谐振子的零点能

振子能量

$$E = \bar{H} = \frac{\overline{p^2}}{2\mu} + \frac{1}{2}\mu\omega^2\overline{x^2}$$

$$\begin{cases} \overline{(\Delta x)^2} = \overline{x^2} - \bar{x}^2 \\ \overline{(\Delta p)^2} = \overline{p^2} - \bar{p}^2 \end{cases} \rightarrow \begin{cases} \overline{x^2} = \overline{(\Delta x)^2} + \bar{x}^2 \\ \overline{p^2} = \overline{(\Delta p)^2} + \bar{p}^2 \end{cases}$$

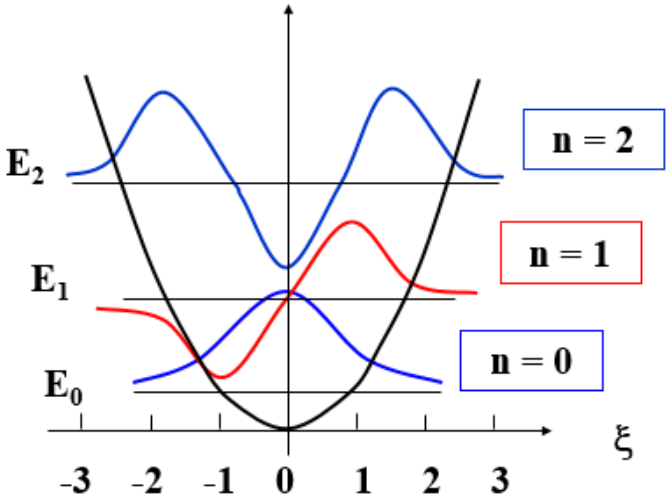


$$\bar{x} = \int_{-\infty}^{\infty} \psi_n^* x \psi_n dx$$

$$= N_n^2 \int_{-\infty}^{\infty} x e^{-\alpha^2 x^2} H_n^2(\alpha x) dx$$

被积函数是 x 的奇函数 $= 0$

测不准关系



$$\overline{p}=\int_{-\infty}^{\infty}\psi_n^{*}\hat{p}\psi_n dx=-i\hbar\int_{-\infty}^{\infty}\psi_n^{*}\frac{\partial}{\partial x}\psi_n dx$$

$$=-i\hbar\psi_n^{*}\psi_n\Big|_{-\infty}^{\infty}+i\hbar\int_{-\infty}^{\infty}\psi_n\frac{\partial}{\partial x}\psi_n^{*}dx$$

∞处$\psi_n=0$

$$=i\hbar\int_{-\infty}^{\infty}\psi_n\frac{\partial}{\partial x}\psi_n^{*}dx$$

ψ_n为实

$$=i\hbar\int_{-\infty}^{\infty}\psi_n^{*}\frac{\partial}{\partial x}\psi_n dx=-\overline{p}$$

$$\therefore\overline{p}=0$$

$$E=\overline{H}=\frac{\overline{p^2}}{2\mu}+\frac{1}{2}\mu\omega^2\overline{x^2}=\frac{\overline{[(\Delta p)^2+\overline{p}^2]}}{2\mu}+\frac{1}{2}\mu\omega^2\overline{[(\Delta x)^2+\overline{x}^2]}$$

$\overline{x}=0,\overline{p}=0$

$$=\frac{\overline{(\Delta p)^2}}{2\mu}+\frac{1}{2}\mu\omega^2\overline{(\Delta x)^2}$$

测不准关系

$$E=\frac{\overline{(\Delta p)^2}}{2\mu}+\frac{1}{2}\mu\omega^2\overline{(\Delta x)^2}$$

$$\overline{(\Delta x)^2}\cdot\overline{(\Delta p_x)^2}\geq\frac{\hbar^2}{4}$$

$$\geq\frac{1}{2\mu}\frac{\hbar^2}{4\overline{(\Delta x)^2}}+\frac{1}{2}\mu\omega^2\overline{(\Delta x)^2}$$

E 的最小值发生在取等号时

$$E=\frac{\hbar^2}{8\mu\overline{(\Delta x)^2}}+\frac{1}{2}\mu\omega^2\overline{(\Delta x)^2}=\frac{\hbar^2}{8\mu y}+\frac{1}{2}\mu\omega^2 y$$

求极值：

$$\frac{\partial E}{\partial y}=-\frac{\hbar^2}{8\mu y^2}+\frac{1}{2}\mu\omega^2=0\quad\Rightarrow\quad y=\pm\frac{\hbar}{2\mu\omega}=\overline{(\Delta x)^2}$$

因均方偏差不能小于零，故取正

$$E=\frac{\hbar^2}{8\mu\left(\frac{\hbar}{2\mu\omega}\right)}+\frac{1}{2}\mu\omega^2\left(\frac{\hbar}{2\mu\omega}\right)=\frac{1}{2}\hbar\omega$$

零点能就是测不准关系所要求的最小能量

测不准关系

例：利用测不准关系证明，在 L_z 本征态 Y_{lm} 下，$\langle L_x \rangle = \langle L_y \rangle = 0$

证：

$$\because [\hat{L}_y, \hat{L}_z] = i\hbar \hat{L}_x \qquad \therefore \overline{(\Delta L_y)^2} \cdot \overline{(\Delta L_z)^2} \geq \frac{\hbar^2}{4} \overline{L_x}^2$$

由于在 L_z 本征态 Y_{lm} 中，测量力学量 L_z 有确定值，所以 L_z 均方偏差必为零，即 $\overline{(\Delta L_z)^2} = 0$

则测不准关系：

$$\overline{(\Delta L_y)^2} \cdot 0 \geq \frac{\hbar^2}{4} \overline{L_x}^2 \Rightarrow 0 \geq \frac{\hbar^2}{4} \overline{L_x}^2$$

平均值的平方为非负数 $\overline{L_x} = 0$

同理：$\overline{L_y} = 0$

测不准关系

角动量分量间的测不准关系

$$\because [\hat{L}_x, \hat{L}_y] = i\hbar \hat{L}_z \qquad \therefore \overline{(\Delta L_x)^2} \cdot \overline{(\Delta L_y)^2} \geq \frac{\hbar^2}{4} \overline{L_z}^2$$

当体系处于$\hat{L}_z$本征态时，

$$\overline{(\Delta L_x)^2} \cdot \overline{(\Delta L_y)^2} \geq \frac{\hbar^2}{4}(m\hbar)^2 = \frac{1}{4} m^2 \hbar^4$$

m=0 时

$$\overline{(\Delta L_x)^2} \cdot \overline{(\Delta L_y)^2} \geq 0$$

$\ell = 0$ 的态 $\mathbf{Y_{00}}$, $\mathbf{L_x}$, $\mathbf{L_y}$, $\mathbf{L_z}$ 同时有确定值

$$\overline{(\Delta L_x)^2} \cdot \overline{(\Delta L_y)^2} = 0$$

测不准关系

例：L^2，L_Z 共同本征态 Y_{lm} 下，求测不准关系：

$$\overline{(\Delta L_x)^2}\cdot\overline{(\Delta L_y)^2}=?$$

$$\overline{(\Delta L_x)^2}=\overline{L_x^{\,2}}-\overline{L_x}^2$$
$$\overline{(\Delta L_y)^2}=\overline{L_y^{\,2}}-\overline{L_y}^2$$

$$\overline{L_x}=0,\overline{L_y}=0$$

$$\overline{(\Delta L_x)^2}=\overline{L_x^{\,2}}$$
$$\overline{(\Delta L_y)^2}=\overline{L_y^{\,2}}$$

$$i\hbar\hat{L}_x^{\,2}=i\hbar\hat{L}_x\hat{L}_x$$

$$\hat{L}_y\hat{L}_z-\hat{L}_z\hat{L}_y=i\hbar\hat{L}_x$$

$$=\hat{L}_y\hat{L}_z\hat{L}_x-\hat{L}_z\hat{L}_y\hat{L}_x$$

$$\hat{L}_x\hat{L}_z-\hat{L}_z\hat{L}_x=-i\hbar\hat{L}_y$$

$$=\hat{L}_y(\hat{L}_x\hat{L}_z+i\hbar\hat{L}_y)-\hat{L}_z\hat{L}_y\hat{L}_x$$

$$=\hat{L}_y\hat{L}_x\hat{L}_z+i\hbar\hat{L}_y^{\,2}-\hat{L}_z\hat{L}_y\hat{L}_x$$

测不准关系

$$i\hbar\hat{L}_x^{\,2} = \hat{L}_y\hat{L}_x\hat{L}_z + i\hbar\hat{L}_y^{\,2} - \hat{L}_z\hat{L}_y\hat{L}_x$$

将上式两边在 $\mathbf{Y_{lm}}$ 态下求平均：

$$i\hbar\int Y_{lm}^*\hat{L}_x^{\,2}Y_{lm}d\Omega = \int Y_{lm}^*\hat{L}_y\hat{L}_x\hat{L}_zY_{lm}d\Omega + i\hbar\int Y_{lm}^*\hat{L}_y^{\,2}Y_{lm}d\Omega - \int Y_{lm}^*\hat{L}_z\hat{L}_y\hat{L}_xY_{lm}d\Omega$$

Lz是厄密算符

$$= \int Y_{lm}^*\hat{L}_y\hat{L}_x\hat{L}_zY_{lm}d\Omega + i\hbar\overline{\hat{L}_y^{\,2}} - \int(\hat{L}_zY_{lm})^*\hat{L}_y\hat{L}_xY_{lm}d\Omega$$

$\hat{L}_zY_{lm} = m\hbar Y_{lm}$

$$= m\hbar\int Y_{lm}^*\hat{L}_y\hat{L}_xY_{lm}d\Omega + i\hbar\overline{\hat{L}_y^{\,2}} - m\hbar\int Y_{lm}^*\hat{L}_y\hat{L}_xY_{lm}d\Omega$$

$$= i\hbar\overline{\hat{L}_y^{\,2}}$$

$$\Rightarrow \overline{L_x^{\,2}} = \overline{\hat{L}_y^{\,2}}$$

测不准关系

$$\overline{\hat{L}_x^{\,2}}+\overline{\hat{L}_y^{\,2}}=\int Y_{lm}^{*}\hat{L}_x^{\,2}Y_{lm}d\Omega+\int Y_{lm}^{*}\hat{L}_y^{\,2}Y_{lm}d\Omega$$

$$=\int Y_{lm}^{*}(\hat{L}_x^{\,2}+\hat{L}_y^{\,2})Y_{lm}d\Omega$$

$$\hat{L}^2=\hat{L}_x^{\,2}+\hat{L}_y^{\,2}+\hat{L}_z^{\,2}$$

$$=\int Y_{lm}^{*}(\hat{L}^2-\hat{L}_z^{\,2})Y_{lm}d\Omega$$

$$=[l(l+1)\hbar^2-m^2\hbar^2]\int Y_{lm}^{*}Y_{lm}d\Omega$$

$$=l(l+1)\hbar^2-m^2\hbar^2$$

$$\because\ \overline{L_x^{\,2}}=\overline{\hat{L}_y^{\,2}}\qquad\therefore\ \overline{\hat{L}_x^{\,2}}=\overline{\hat{L}_y^{\,2}}=\frac{1}{2}[l(l+1)-m^2]\hbar^2$$

$$\overline{(\Delta L_x)^2}=\overline{L_x^{\,2}}-\overline{L_x}^{\,2}=\overline{L_x^{\,2}}$$

$$\overline{(\Delta L_y)^2}=\overline{L_y^{\,2}}-\overline{L_y}^{\,2}=\overline{L_y^{\,2}}$$

测不准关系：

$$\overline{(\Delta L_x)^2}\cdot\overline{(\Delta L_y)^2}=\frac{1}{4}[(l(l+1)-m^2]^2\hbar^4$$

测不准关系

例：L^2，L_Z 共同本征态 Y_{lm} 下，求测不准关系（利用角动量升降算符）：

$$\overline{(\Delta L_x)^2}\cdot\overline{(\Delta L_y)^2}=?$$

$$\overline{(\Delta L_x)^2}=\overline{L_x^{\,2}}-\overline{L_x}^2$$
$$\overline{(\Delta L_y)^2}=\overline{L_y^{\,2}}-\overline{L_y}^2$$

$$\overline{L_x}=0,\overline{L_y}=0$$

$$\overline{(\Delta L_x)^2}=\overline{L_x^{\,2}}$$
$$\overline{(\Delta L_y)^2}=\overline{L_y^{\,2}}$$

测不准关系

$$\overline{\hat{L}_x^{\,2}} = \int Y_{lm}^* \hat{L}_x^{\,2} Y_{lm} d\Omega$$

$\hat{L}_+ = \hat{L}_x + i\hat{L}_y, \hat{L}_- = \hat{L}_x - i\hat{L}_y$

$$= \int Y_{lm}^* \frac{1}{2}(\hat{L}_+ + \hat{L}_-)\frac{1}{2}(\hat{L}_+ + \hat{L}_-) Y_{lm} d\Omega$$

$$= \frac{1}{4}\int Y_{lm}^* (\hat{L}_+^{\,2} + 2\hat{L}_+\hat{L}_- + \hat{L}_-^{\,2}) Y_{lm} d\Omega$$

$$= \frac{1}{4}\int Y_{lm}^* (\hat{L}_+^{\,2}) Y_{lm} d\Omega + \frac{1}{4}\int Y_{lm}^* (2\hat{L}_+\hat{L}_-) Y_{lm} d\Omega + \frac{1}{4}\int Y_{lm}^* (\hat{L}_-^{\,2}) Y_{lm} d\Omega$$

$\hat{L}_\pm Y_{lm} = \hbar\sqrt{l(l+1) - m(m\pm 1)}\, Y_{l,m\pm 1}$

$$= 0 + \frac{1}{4}\int Y_{lm}^* (2\hat{L}_+\hat{L}_-) Y_{lm} d\Omega + 0$$

$$= \frac{1}{2}\int Y_{lm}^* (\hat{L}_+\hat{L}_-) Y_{lm} d\Omega = \frac{1}{2}\hbar\sqrt{l(l+1) - m(m-1)}\int Y_{lm}^* (\hat{L}_+) Y_{l,m-1} d\Omega$$

$$= \frac{1}{2}\hbar\sqrt{l(l+1) - m(m-1)}\,\hbar\sqrt{l(l+1) - (m-1)m}\int Y_{lm}^* Y_{l,m} d\Omega$$

$$= \frac{1}{2}[l(l+1) - m(m-1)]\hbar^2$$

同理：　$\overline{\hat{L}_y^{\,2}} = \frac{1}{2}[l(l+1) - m(m-1)]\hbar^2$

测不准关系

$$\overline{(\Delta L_x)^2} = \overline{L_x^2} - \overline{L_x}^2 = \overline{L_x^2}$$
$$\overline{(\Delta L_y)^2} = \overline{L_y^2} - \overline{L_y}^2 = \overline{L_y^2}$$

$$\overline{\hat{L}_x^2} = \overline{\hat{L}_y^2} = \frac{1}{2}\hbar^2[l(l+1) - m(m-1)]$$

$$\overline{(\Delta L_x)^2} \cdot \overline{(\Delta L_y)^2} = \overline{L_x^2} \cdot \overline{L_y^2}$$

$$= \left(\frac{1}{2}[l(l+1) - m(m-1)]\hbar^2\right)^2$$

$$\overline{(\Delta L_x)^2} \cdot \overline{(\Delta L_y)^2} = \frac{1}{4}[l(l+1) - m(m-1)]^2\hbar^4$$

第一种方法：

$$\overline{(\Delta L_x)^2} \cdot \overline{(\Delta L_y)^2} = \frac{1}{4}[(l(l+1) - m^2]^2\hbar^4$$

两种方法计算的结果不一致



测不准关系

$$\overline{\hat{L}_x^{\,2}} = \int Y_{lm}^* \hat{L}_x^{\,2} Y_{lm} d\Omega$$

$\hat{L}_+ = \hat{L}_x + i\hat{L}_y, \hat{L}_- = \hat{L}_x - i\hat{L}_y$

$$= \int Y_{lm}^* \frac{1}{2}(\hat{L}_+ + \hat{L}_-)\frac{1}{2}(\hat{L}_+ + \hat{L}_-) Y_{lm} d\Omega$$

$$= \frac{1}{4}\int Y_{lm}^* (\hat{L}_+^{\,2} + \hat{L}_+\hat{L}_- + \hat{L}_-\hat{L}_+ + \hat{L}_-^{\,2}) Y_{lm} d\Omega$$

$$= \frac{1}{4}\int Y_{lm}^* (\hat{L}_+^{\,2}) Y_{lm} d\Omega + \frac{1}{4}\int Y_{lm}^* (\hat{L}_+\hat{L}_-) Y_{lm} d\Omega + \frac{1}{4}\int Y_{lm}^* (\hat{L}_-\hat{L}_+) Y_{lm} d\Omega + \frac{1}{4}\int Y_{lm}^* (\hat{L}_-^{\,2}) Y_{lm} d\Omega$$

$\hat{L}_\pm Y_{lm} = \hbar\sqrt{l(l+1) - m(m\pm 1)}\, Y_{l,m\pm 1}$

$$= 0 + \frac{1}{4}\int Y_{lm}^* (\hat{L}_+\hat{L}_-) Y_{lm} d\Omega + \frac{1}{4}\int Y_{lm}^* (\hat{L}_-\hat{L}_+) Y_{lm} d\Omega + 0$$

$$= \frac{1}{4}\int Y_{lm}^* (\hat{L}_+\hat{L}_-) Y_{lm} d\Omega + \frac{1}{4}\int Y_{lm}^* (\hat{L}_-\hat{L}_+) Y_{lm} d\Omega$$

$$= \frac{1}{4}\hbar\sqrt{l(l+1) - m(m-1)}\int Y_{lm}^* (\hat{L}_+) Y_{l,m-1} d\Omega + \frac{1}{4}\hbar\sqrt{l(l+1) - m(m+1)}\int Y_{lm}^* (\hat{L}_-) Y_{l,m+1} d\Omega$$

$$= \frac{1}{4}\hbar\sqrt{l(l+1) - m(m-1)}\hbar\sqrt{l(l+1) - (m-1)m} + \frac{1}{4}\hbar\sqrt{l(l+1) - m(m+1)}\hbar\sqrt{l(l+1) - (m+1)m}$$

$$= \frac{1}{4}[l(l+1) - m(m-1)]\hbar^2 + \frac{1}{4}[l(l+1) - m(m+1)]\hbar^2 = \frac{1}{2}[l(l+1) - m^2]\hbar^2$$

测不准关系

$$\overline{(\Delta L_x)^2} = \overline{L_x^2} - \overline{L_x}^2 = \overline{L_x^2}$$
$$\overline{(\Delta L_y)^2} = \overline{L_y^2} - \overline{L_y}^2 = \overline{L_y^2}$$

$$\overline{\hat{L}_x^2} = \overline{\hat{L}_y^2} = \frac{1}{2}\hbar^2[l(l+1) - m(m-1)]$$

$$\overline{(\Delta L_x)^2} \cdot \overline{(\Delta L_y)^2} = \overline{L_x^2} \cdot \overline{L_y^2}$$
$$= \left(\frac{1}{2}[l(l+1) - m^2]\hbar^2\right)^2$$

$$\overline{(\Delta L_x)^2} \cdot \overline{(\Delta L_y)^2} = \frac{1}{4}[l(l+1) - m^2]^2\hbar^4$$

第一种方法：
$$\overline{(\Delta L_x)^2} \cdot \overline{(\Delta L_y)^2} = \frac{1}{4}[(l(l+1) - m^2]^2\hbar^4$$

测不准关系

如何理解下面两式：

$$\overline{(\Delta L_x)^2}\cdot\overline{(\Delta L_y)^2}=\frac{1}{4}[(l(l+1)-m^2]^2\hbar^4$$

$$\overline{(\Delta L_x)^2}\cdot\overline{(\Delta L_y)^2}\geq\frac{1}{4}m^2\hbar^4$$

考察下式的最小值：

$$\frac{1}{4}[(l(l+1)-m^2]^2\hbar^4$$

$m=\pm l$

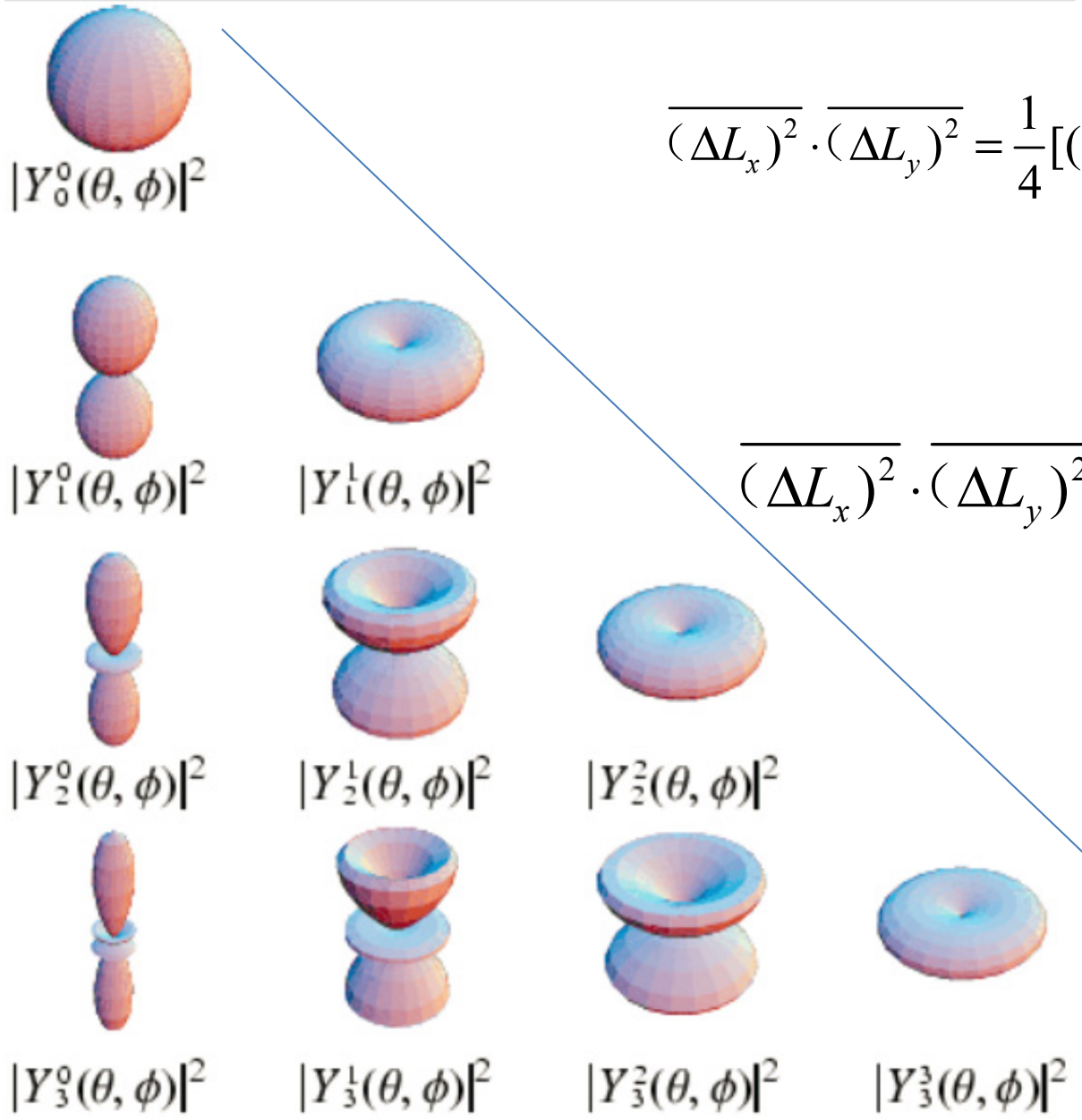
$$=\frac{1}{4}[(l(l+1)-l^2]^2\hbar^4$$

$$=\frac{1}{4}l^2\hbar^4$$

$m=\pm l$

$$=\frac{1}{4}m^2\hbar^4$$

测不准关系



$$\overline{(\Delta L_x)^2}\cdot\overline{(\Delta L_y)^2}=\frac{1}{4}[(l(l+1)-m^2]^2\hbar^4$$

$$\overline{(\Delta L_x)^2}\cdot\overline{(\Delta L_y)^2}\geq\frac{1}{4}m^2\hbar^4$$

力学量守恒

$$\bar{F}=\int \Psi^*(x,t)\hat{F}\Psi(x,t)dx$$

$\Psi(x,t)$是时间的函数，$\hat{F}$也可能显含时间

$$\frac{d\bar{F}}{dt}=\int \Psi^*\frac{d\hat{F}}{dt}\Psi dx+\int \frac{d\Psi^*}{dt}\hat{F}\Psi dx+\int \Psi^*\hat{F}\frac{d\Psi}{dt}dx$$

利用Schrödinger方程

$$=\int \Psi^*\frac{d\hat{F}}{dt}\Psi dx+\int\left[-\frac{1}{i\hbar}\left(\hat{H}\Psi\right)^*\right]\hat{F}\Psi dx+\int \Psi^*\hat{F}\left(\frac{1}{i\hbar}\hat{H}\Psi\right)dx$$

整理

$$=\int \Psi^*\frac{d\hat{F}}{dt}\Psi dx-\frac{1}{i\hbar}\int\left(\hat{H}\Psi\right)^*\hat{F}\Psi dx+\frac{1}{i\hbar}\int \Psi^*\hat{F}\left(\hat{H}\Psi\right)dx$$

H 是厄密算符

$$=\int \Psi^*\frac{d\hat{F}}{dt}\Psi dx-\frac{1}{i\hbar}\int \Psi^*\hat{H}\hat{F}\Psi dx+\frac{1}{i\hbar}\int \Psi^*\hat{F}\hat{H}\Psi dx$$

整理、利用对易括号

$$=\int \Psi^*\frac{d\hat{F}}{dt}\Psi dx+\frac{1}{i\hbar}\int \Psi^*[\hat{F},\hat{H}]\Psi dx$$

利用期望值定义

$$=\overline{\frac{d\hat{F}}{dt}}+\frac{1}{i\hbar}\overline{[\hat{F},\hat{H}]}$$

力学量守恒

$$\frac{d\bar{F}}{dt}=\overline{\frac{d\hat{F}}{dt}}+\frac{1}{i\hbar}\overline{[\hat{F},\hat{H}]}$$

若$\hat{F}$不显含时间，例如：x, p_x, L_x

$$\frac{d\bar{F}}{dt}=\frac{1}{i\hbar}\overline{[\hat{F},\hat{H}]}$$

若$\hat{F}$既不显含时间，又与$\hat{H}$对易

$$\frac{d\bar{F}}{dt}=0$$

F的期望值不随时间改变。该力学量F为运动恒量，或称F在运动中守恒

力学量守恒

（1）自由粒子的动量

当粒子不受外力作用时，其哈密顿算符为

$$\hat{H}=\frac{\hat{p}^2}{2m}$$

考察

$$\frac{d\overline{p}}{dt}=\overline{\frac{d\hat{p}}{dt}}+\frac{1}{i\hbar}\overline{[\hat{p},\hat{H}]}=\frac{1}{i\hbar}\overline{[\hat{p},\hat{H}]}=0$$

自由粒子的动量是运动恒量。这就是量子力学中的动量守恒定律。

力学量守恒

（2）中心力场中运动粒子的角动量

粒子在势函数为V(r)的中心力场中运动时，哈密顿算符为

$$\hat{H}=-\frac{\hbar^2}{2mr^2}\frac{\partial}{\partial r}(r^2\frac{\partial}{\partial r})+\frac{\hat{L}^2}{2mr^2}+V(r)$$

考察

$$\frac{d\overline{\hat{L}^2}}{dt}=\overline{\frac{d\hat{L}^2}{dt}}+\frac{1}{i\hbar}\overline{[\hat{L}^2,\hat{H}]}=\frac{1}{i\hbar}\overline{[\hat{L}^2,\hat{H}]}=0$$

$$\hat{L}^2=-\hbar^2[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial}{\partial\theta})+\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}]$$

$\hat{L}^2$只与θ、φ有关，与r无关

$\hat{L}^2$与$\hat{H}$对易

$$\hat{L}_x=i\hbar[\sin\varphi\frac{\partial}{\partial\theta}+\cot\theta\cos\varphi\frac{\partial}{\partial\varphi}],\hat{L}_y=-i\hbar[\cos\varphi\frac{\partial}{\partial\theta}+\cot\theta\sin\varphi\frac{\partial}{\partial\varphi}],\hat{L}_z=-i\hbar\frac{\partial}{\partial\varphi}$$

$$\frac{d\overline{\hat{L}^2}}{dt}=0,\frac{d\overline{\hat{L}_x}}{dt}=0,\frac{d\overline{\hat{L}_y}}{dt}=0,\frac{d\overline{\hat{L}_z}}{dt}=0$$

中心力场中运动的粒子，其角动量平方和角动量分量都是运动恒量，即量子力学中的角动量守恒定律。

力学量守恒

（3）哈密顿不显含时间的体系的能量

若体系的哈密顿算符 H 不是时间的显函数，则

$$\frac{d\bar{H}}{dt}=\overline{\frac{d\hat{H}}{dt}}+\frac{1}{i\hbar}\overline{[\hat{H},\hat{H}]}=\frac{1}{i\hbar}\overline{[\hat{H},\hat{H}]}=0$$

体系的能量是运动恒量。这就是量子力学中的能量守恒定律。

如一维无限深势阱中的粒子、线性谐振子、氢原子等，它们的哈密顿不显含时间，所以它们的能量都是运动恒量。

力学量守恒

（4）哈密顿对空间反演不变时的宇称

把一个函数的所有坐标变量改变符号的运算称为空间反演，若定义相应的算符为P（称为宇称算符），则

$$\hat{P}\Psi(x,t)=\Psi(-x,t)$$

考察P的本征值，将P同时作用于上式，可得：

$$\hat{P}\hat{P}\Psi(x,t)=\hat{P}\Psi(-x,t)=\Psi(x,t)$$

P^2的本征值为1，所以P的本征值为1或-1，即：

$\hat{P}\Psi_1=\Psi_1$　本征值为1的本征函数空间反演后不变，具有偶宇称

$\hat{P}\Psi_2=-\Psi_2$　本征值为-1的本征函数空间反演后反号，具有奇宇称

力学量守恒

若体系的哈密顿算符 H 空间反演后保持不变

$$\hat{H}(x)=\hat{H}(-x)$$

考察 H 与宇称算符 P 的对易

$$\hat{P}\hat{H}(x)\Psi(x,t)=\hat{H}(-x)\Psi(-x,t)=\hat{H}(x)\Psi(-x,t)=\hat{H}(x)\hat{P}\Psi(x,t)$$

考察

$$\frac{d\bar{P}}{dt}=\overline{\frac{d\hat{P}}{dt}}+\frac{1}{i\hbar}\overline{[\hat{P},\hat{H}]}=\frac{1}{i\hbar}\overline{[\hat{P},\hat{H}]}=0$$

宇称是运动恒量。

H 与宇称算符 P 可以有共同本征函数，即体系的能量本征函数可以有确定的宇称，而且不随时间改变。这即是量子力学中的宇称守恒定律。

力学量守恒

量子力学中一个不可观察量的对称性（不变性）导致一个可观察量的守恒性：

空间平移对称性 ⟹ 动量守恒

空间旋转对称性 ⟹ 角动量守恒

时间平移对称性 ⟹ 能量守恒

空间反演对称性 ⟹ 宇称守恒

作业　曾谨言《量子力学》卷I（第五版）

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